# FUNCTIONAL HERBAL TEA INCORPORATING CORDYCEPS AND BIOACTIVE BOTANICALS TARGETING DIABETIC ENZYMES: A BIOCHEMICAL APPROACH

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## INTRODUCTION:

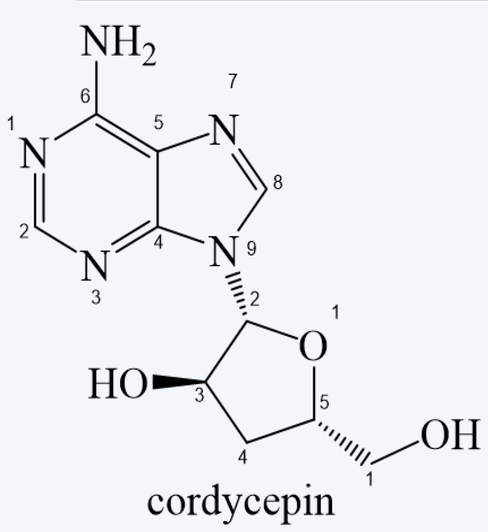
T2DM is linked to insulin resistance and postprandial hyperglycemia. Phytochemicals from Cordyceps militaris and Gymnema sylvestre offer antioxidant and enzyme-inhibitory effects related to glycemic control.



*Cordyceps militaris*



*Gymnema sylvestre*

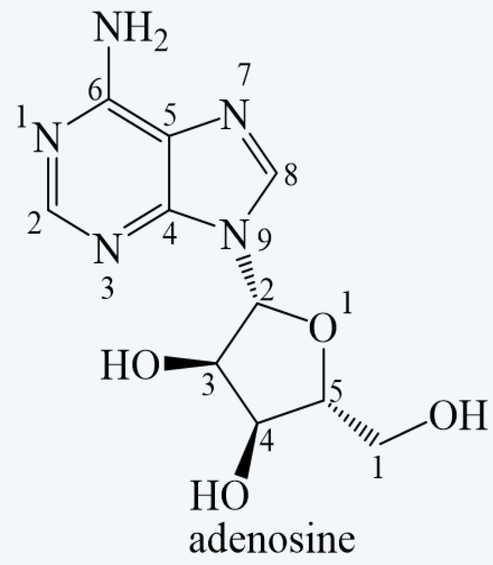


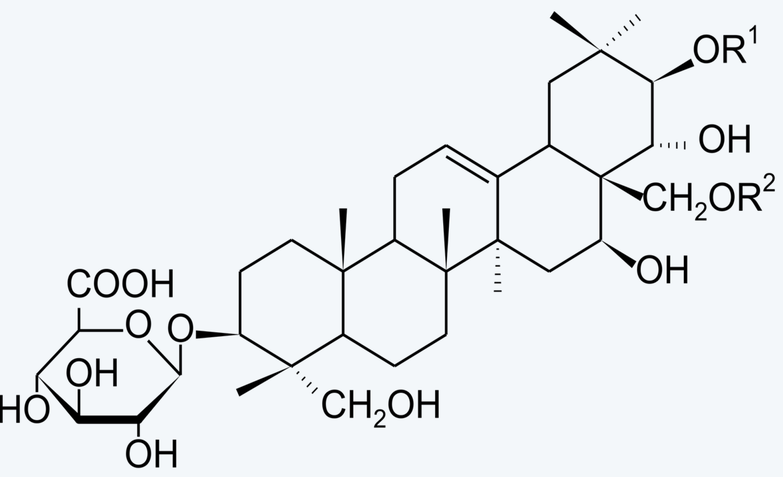

## METHODS:

- **Materials:** Cordyceps militaris (fruiting body) + Gymnema sylvestre (dried leaves).
- **Solvents:** cold water, boiling water, 70% EtOH, 97% EtOH.
- **Ratios tested:** 1:1 → 3:2 (Cordyceps:Gymnema).
- **Assays:** DPPH; $IC_{50}$ for α-amylase & α-glucosidase.

## RESULTS:

- Best ratio: 3:2, with strong enzyme inhibition:
  - α-amylase $IC_{50}$: 171.8 ± 2.1 µg/mL
  - α-glucosidase $IC_{50}$: 181.7 ± 2.3 µg/mL
- DPPH scavenging: 56.4 ± 2.1% (moderate).
- Optimal extraction: 70% EtOH (70°C) or boiling water.
- 97% EtOH and cold water yielded lower activity.
- Differences between ratios/solvents were statistically significant ($p < 0.05$).




gymnemic acid

## CONCLUSION:

- The 3:2 Cordyceps–Gymnema blend shows strong in-vitro antidiabetic activity through dual inhibition of α-amylase and α-glucosidase. Extraction with 70% ethanol (70°C) or boiling water is optimal for safe, bioactive herbal tea development.

## RECOMMENDATIONS:

Further work should evaluate in-vivo efficacy, bioavailability, product stability, and sensory acceptance, alongside standardizing key actives (cordycepin, gymnemic acids). The formulation shows strong potential for development into OTC functional beverages for prediabetic and diabetic users.